

Assignment:-

Build the Backend for a Personal Book Library

Objective

Develop the backend for a **Personal Book Library** application using **Node.js**, **Express.js**, and **MongoDB**. The goal of this assignment is to practice creating RESTful APIs, working with MongoDB, handling HTTP requests and responses, and organizing backend code following a modular structure.

Features to Implement

1. Create a Book

Implement an API endpoint that allows users to add a new book to the library.

Each book should contain the following fields:

- **Title**
 - **Author**
 - **Genre**
 - **Rating** (1–5)
 - **Status** (Read / Unread)
 - **Description**
-

2. Get All Books

Create an API endpoint to fetch and return all books stored in the database.

3. Update a Book

Create an API endpoint that allows users to update the details of an existing book.

Users should be able to update any field of a book.

4. Delete a Book

Create an API endpoint that allows users to remove a book from the library.

5. Search Books

Implement a search feature that allows users to search books by:

- **Title**
- **Author**

The search should support partial matches.

API Endpoints

Method	Endpoint	Description
GET	<code>/api/books</code>	Get all books
POST	<code>/api/books</code>	Create a new book
PUT	<code>/api/books/:id</code>	Update an existing book
DELETE	<code>/api/books/:id</code>	Delete a book

Validation Rules

- Title is required.
- Author is required.

- Genre is required.
- Rating must be between **1 and 5**.
- Status should only accept **Read** or **Unread**.
- Return appropriate error messages for invalid input.

Suggested Folder Structure

None

```
server/  
|  
├─ config/  
|   └─ db.js  
|  
├─ controllers/  
|   └─ bookController.js  
|  
├─ models/  
|   └─ Book.js  
|  
├─ routes/  
|   └─ bookRoutes.js  
|  
├─ .env  
├─ server.js  
├─ package.json  
└─ .gitignore
```